Chapter 4: Alternating Current

Electromagnetic Induction?

Faraday's Laws of Electromagnetic Induction

Faraday's Law of Electromagnetic Induction is the basic law of electromagnetism that is used to explain the working of various equipment that includes an electric motor, electric generator, etc. Faraday's law was given by an English scientist Michael Faraday in 1831. According to Faraday's Law of Electromagnetic Induction, the induced current in the circuit is directly proportional to the rate of change of Magnetic Flux.

The basic law of electromagnetic induction predicting how a magnetic field interacts with an electric circuit to produce the electromotive force (EMF) is called Faraday's Law. And this phenomenon of producing the electromotive force in the electric circuit by the interaction of the magnetic field is called Electromagnetic Induction.

Faraday's Laws of Electromagnetic Induction

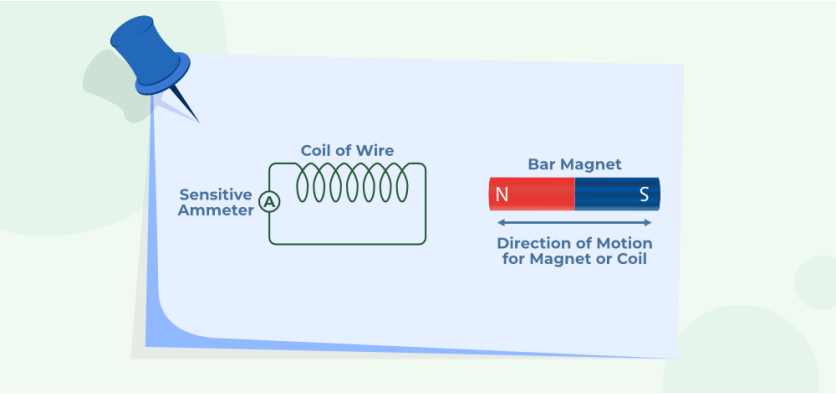
Faraday has provided two laws that are the basis of modern electromagnetism. The laws are discussed below:

- Faraday's First Law of Electromagnetic Induction
- Faraday's Second Law of Electromagnetic Induction

Faraday's First Law of Electromagnetic Induction

According to Faraday's First Law of Electromagnetic Induction, "When the Magnetic Flux linked with closed-circuit changes, an EMF is induced in it which lasts only as long as the change in magnetic flux is taking place. If the circuit is closed then current also gets induced inside the circuit which is called 'Induced current". Changing the magnetic field changes the induced current in the circuit.

The image given below shows the deflection in the coil according to the law of Electromagnetic Induction.



Magnetic Fields Can be changed by,

- *Moving a bar magnet towards or away from the coil.*
- *Moving the coil into the magnetic field or outside the magnetic field.*

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- *Rotating the coil relative to the magnet.*
- *Changing the area of a coil placed in the magnetic field.*

Faraday's Second Law: The Role of Magnetic Flux

According to Faraday's Second Law of Electromagnetic Induction, "The magnitude of the induced emf is equal to the rate of change of magnetic flux linked with the coil."

The second law of electromagnetic induction according to Faraday claims that, the induced emf in a coil is directly proportional to the rate of change of flux linked to the coil. Mathematically, this can be expressed as:

$E = -N d\Phi_B/dt$ [Equation 1]
Where,

- E is the induced electromotive force or voltage that is measured in volt,
- N is the number of turns present in the coil,
- ΦB is the magnetic flux taken in Weber (Wb), and
- 't' is the time taken in seconds.

Hence it can be observed from equation (1) that fast

Faraday's Experiments

Faraday has performed three experiments that form the basis of electromagnetic induction.

Experiment 1

In this experiment, Faraday took a circular coil and connected it with a galvanometer and now he takes a strong bar magnet. When the north pole of the bar magnet is moved towards the coil, the galvanometer showed deflection to the right side of the zero mark in the galvanometer. When the magnet is moved away from the coil again it showed deflection but in the opposite direction. Similarly, the experiment is done with the south pole of the bar magnet, again the deflection is observed but opposite to the direction shown by the north pole of the bar magnet. When the magnet is held stationary near the coil, no deflection is observed in the galvanometer.

Conclusion: As the magnet is moved closer to the coil the magnetic flux increases hence, an induced current setup in the coil in one direction. When the magnet is moved away from the coil, the magnetic flux decreases, hence an induced current set up in the coil in the opposite direction. When the magnet is held stationary near the coil, there is no change in the magnetic flux.

Experiment 2

As you can see from the figure below. Two coils primary (p) and secondary (s), are wound on cylindrical support. The primary coil is connected to a key, a rheostat, and a

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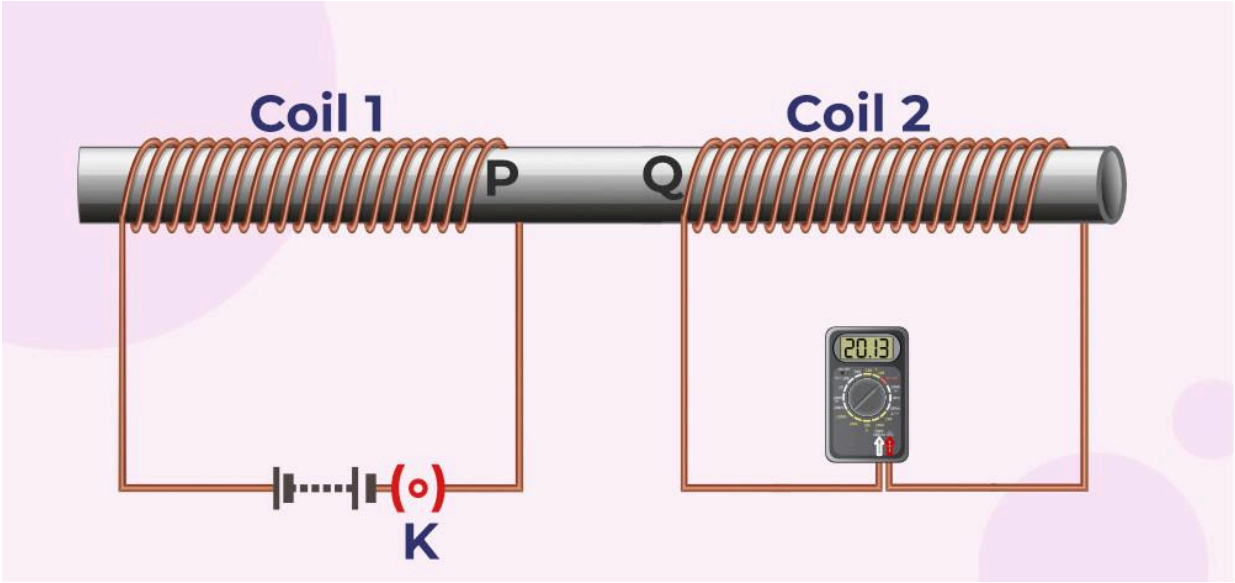
battery. The secondary is connected with a galvanometer. When the key is pressed in the primary coil the galvanometer shows deflection in one direction. When the key is released, it again shows deflection but in the opposite direction. When the key is kept pressed steady current flows through the primary coils, and the galvanometer does not show any deflection. When the current in the primary coil is increased with the help of the rheostat, the induced current flows in the secondary coil in the same direction as that of the primary coil.

The image given below shows the setup of Faraday's Experiment.



Lenz's Law Definition

Lenz's Law is named after the German physicist "Emil Lenz ", who formulated it in 1834. According to Lenz Law, "the direction of induced current in a circuit is such that it opposes the change in magnetic flux produced." It is a scientific law that specifies the direction of induced current but states nothing about its magnitude.

According to Faraday's Second Law of Electromagnetic Induction,

$$E = -N(d\emptyset/dt)$$

Here, the negative sign indicates that the direction of induced emf is such that it opposes the change in magnetic flux which is in accordance with Lenz's law

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Electromagnetic induction

Electromagnetic induction was discovered by Michael Faraday in the 1830s, and it describes the process by which a conductor placed in a changing magnetic field causes a voltage to be produced in the conductor.

Self induction

Self-induction is the phenomenon in which a change in electric current in a circuit induces an electromotive force (EMF) in the same circuit. Consider a simple coil of wire. When electric current flows through the coil, a magnetic field is created around it. If the current changes with time, the magnetic field associated with the coil also changes. This changing magnetic field induces an EMF in the same coil that opposes the change in current through it.

Mutual induction

Mutual induction, on the other hand, is the property of two electric circuits whereby any change in current in the first circuit induces an EMF in the second circuit. If you have two coils such that a change in current in one coil will induce a voltage in the other coil, then these coils are said to have mutual induction.

Self Inductance

The induction of electromotive force due to changes in the flowing current or the coil or the conductor's magnetic flux is defined as self-inductance. The magnetic flux at any point inside the circuit is directly proportional to the current flowing through it. Self-inductance is denoted by L (self-inductance coefficient). The self-inductance of a conductor depends on the cross-section area, the total number of turns in the coil, and the coil material's permeability. The following is the relation:

$\Phi \alpha I$

$\varnothing = LI$

$L = \Phi / I$

The rate of change of the magnetic flux can be stated as:

$e = -(d\varnothing)/dt$

$e = -L$

Therefore:

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$-L = dI/dt$

Here, ∅ stands for magnetic flux.

L is self-induced.

I stand for the current in amperes.

Mutual inductance

When the arrangement of coils is such that the current flowing through one coil changes, an emf is induced in the other coil. These coils are said to have a mutual inductance between them. Mutual inductance is denoted by M and is also measured in Henry.

The setup is created such that there are two coils, primary and secondary. A key and a battery are connected to the primary coil, and a galvanometer is attached to the secondary. When the magnetic flux of the two coils changes, an electromotive force, opposing in nature, is produced all across the coils. This is called mutual inductance. The following is the relation:

$\varnothing = MI$

Here, M is the mutual inductance coefficient, and I is the change in current.

$e = -d\varnothing/dt$

$e = -M$

Therefore:

$-M = dI/dt$

M stands for the mutual inductance coefficient.

I denote current in Ampere.

The formula for mutual inductance is used in various calculations and derivations of several quantities related to it. The formula is:

$M = \mu_0\mu_r N_1 N_2 A / L$

μ_0 is the free space's permeability.

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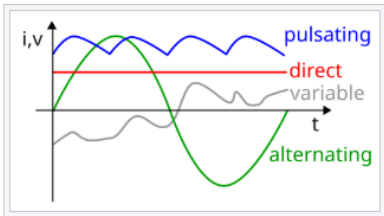
μ_r is the relative permeability of the used iron core.

A stands for the cross-section area in m².

L is the coil's length in m, and N stands for the total number of turns in the coil.

Alternating current (**AC**) is an electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction. Alternating current is the form in which electric power is delivered to businesses and residences, and it is the form of electrical energy that consumers typically use when they plug kitchen appliances, televisions, fans and electric lamps into a wall socket. The abbreviations *AC* and *DC* are often used to mean simply *alternating* and *direct*, respectively, as when they modify *current* or *voltage*.[1][2]



Alternating current (green curve). The horizontal axis measures time (it

What is an RL Circuit?

An RL circuit (also known as an RL filter or RL network) is defined as an electrical circuit consisting of the passive circuit elements of a resistor (R) and an inductor (L) connected together, driven by a voltage source or current source.

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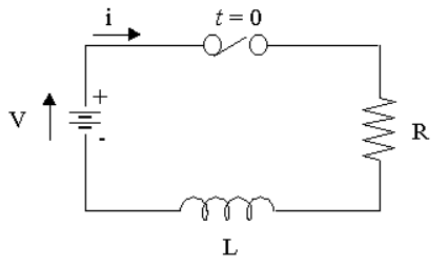
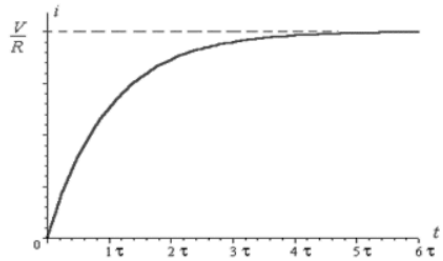
Expression for Current flowing in Series RL Circuit





Consider a circuit in which resistance is connected in series with inductor and voltage source of V volts, is applied across it. Initially the switch is open. Let us say at time 't' we close the switch and the current 'i' starts flowing in the circuit but it does not attains its maximum value rapidly due to the presence of inductor in the circuit as we know inductor has a property to oppose the change in the current flowing through it.

Apply Kirchhoff's voltage law in the above series RL circuit,

$$V - iR - L\frac{di}{dt} = 0$$

Rearranging the above equation,

$$V - iR = L\frac{di}{dt}$$

$$\frac{dt}{L} = \frac{di}{V - iR}$$

Integrating both sides, we get,

$$\int_0^t \frac{dt}{L} = \int_0^i \frac{di}{V - iR}$$

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Now integrate right hand side by using substitution method,

$$Let\ z = V - iR\ or\ \frac{dz}{di} = -R$$

Substituting the values we get,

$$or\ \frac{t}{L} = -\frac{1}{R}\int \frac{dz}{z}$$

We know that integration of,

$$\frac{1}{z} = ln(z)$$

So we get,

$$-\frac{Rt}{L} = ln(V - iR)_0^i$$

By applying limits we get,

$$-\frac{Rt}{L} = ln(V - iR) - ln(V)$$

Simplifying again,

$$-\frac{Rt}{L} = ln\frac{V - iR}{V}$$

Taking antilog on both sides,

$$e^{-\frac{Rt}{L}} = e^{ln\frac{V-iR}{V}}$$

We know that $e^{\ln x} = x$, so we get,

$$e^{-\frac{Rt}{L}} = \frac{V - iR}{V}$$

Moving the term containing 'i' on one side we get,

$$i = \frac{V(1 - e^{-\frac{Rt}{L}})}{R}$$

The term L/R is called the Time Constant (τ) of the **RL series circuit**. It is defined as the time taken for the current to reach its maximum steady state value. The term V/R represents the final steady state value of the current in the circuit.

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Fig. 7.12 Mutual induction.

7.8. Energy Stored in an Inductor

If the current through the inductor is increased from zero to a steady value I, an *emf* is generated which acts to opposite the increase in the current. Work must be done against this *emf* by the voltage source in order to establish the current in the inductor. Let dW be the work done in establishing a current I in the coil in time dt is

$$dW = P\,dt \qquad [\text{ power} = \text{voltage} \times \text{current}]$$

Then, $\quad dW = -\varepsilon\, i\, dt$, where ε is the induced *emf*, $\varepsilon = -L\dfrac{di}{dt}$

To find the total work W done in establishing the final current I in the inductor, we must integrate the above expression. Thus,

$$W = \int_0^W dW = \int_0^I -\varepsilon\, i\, dt$$

$$= \int_0^I Li\frac{di}{dt}\,dt$$

$$= \int_0^I L\,i\,di$$

$$= \frac{1}{2}LI^2$$

Energy stored in the magnetic field, $W = \dfrac{1}{2}LI^2$

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